

You “Claire” Wu

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EDUCATION

Texas A&M University, College of Engineering

College Station, Texas

Bachelor of Science in Computer Science, Minors in Statistics and AI in Business

May 2027

Relevant coursework: Data Structures and Algorithms, Software Engineering, Machine Learning, ML in Business

EXPERIENCE

Dream Lab

College Station, Texas

Undergraduate Researcher

January 2026 – Present

- Built AnnoMath, an AI-powered multimodal math tutoring system mirroring human tutoring by annotating handwritten student work, with a team of researchers; user study showed significant reductions in visual clutter and increased engagement (explanations rated significantly more concise, $p = 0.0021$)
- Implemented multiple annotation types (TypeScript, React, tldraw) covering ~22% of annotation occurrences, identified via frequency analysis of 30 openly coded instructional videos
- Developed React/TypeScript logic to render a baseline (non-annotated) condition for the user study

Aggies Create

College Station, Texas

Team Lead and Consultant for AksaOils

August 2025 – December 2025

- Led team through data-access constraints by training and validating a Random Forest model on simulated order data (98% accuracy in internal testing), establishing technical feasibility for future deployment
- Programmed a personalized product recommendation quiz (HTML, CSS, JavaScript) designed for Shopify integration
- Directed rebranding and workflow, aligning technical components with brand identity under a compressed timeline

Signify Health

College Station, Texas

Software Engineering Consultant

September 2024 – May 2025

- Developed Signify Hex, a full-stack data visualization tool (React/TypeScript, Uber H3, Go, PostgreSQL) connecting medical providers with patients and increasing provider efficiency, through Aggies Create
- Implemented Redis-based WebSocket change tracking to signal frontend data refresh, optimizing performance
- Created travel radius overlays using Mapbox and Uber H3 to show provider reachability by location
- Enhanced UX by adding hover interactions, dark mode, and dynamic data fetching via gRPC
- Presented live demos to Signify Health software engineering leadership and awarded 1st place among consultancy projects at the Aggies Create Innovation Expo (Spring 2025)

PROJECTS

Pearl Tea POS System | AWS, Java, JavaFX, PostgreSQL, Node.js/Express

College Station, Texas

Full Stack Developer

September 2025 – December 2025

- Built a boba shop POS with customer and manager views for inventory and order management using JavaFX and a PostgreSQL database hosted on AWS, following the Waterfall model
- Migrated system to web application under Agile, built a RESTful Node.js/Express API and led deployment on Render with secure authenticated access
- Integrated Deepgram's Text-to-Speech (TTS) API and built a drink recommendation system based on weather to enhance accessibility and user experience
- Resolved incomplete translation output (caused by multiple simultaneous API requests) by implementing a multi-layer caching system (hardcoded dictionary + memory cache) to reduce redundant calls and improve reliability
- Implemented dynamic database-driven menu rendering and customer kiosk flow for checkout and tipping

SKILLS & LANGUAGES

Languages: Python, TypeScript, JavaScript, Java, C++, Go, R, HTML/CSS, Mandarin Chinese (fluent)

Web, ML & Graphics: React, Next.js, React Three Fiber/Three.js, Pandas, Node.js/Express, PyTorch, Scikit-learn, XGBoost, Bootstrap, NumPy, Matplotlib, Seaborn, Tailwind CSS, OpenGL, GLSL, Selenium, gRPC, JavaFX, tldraw

Databases & Tools: PostgreSQL, AWS, Redis, Mapbox, Uber H3, Deepgram, Azure Translator, il8n, Docker, Jira, Render, Figma, Git, GitHub